

Notes: Kline and Moretti (QJE, 2014)

Local Economic Development, Agglomeration Economies, and the Big Push: 100 Years of Evidence from the Tennessee Valley Authority

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1 Big picture

- **Question.** Can a large place-based development policy permanently transform a poor region, and can its local gains translate into national welfare gains rather than just moving jobs across space?
- **Policy studied.** The paper studies the Tennessee Valley Authority (TVA), created in 1933 to modernize the Tennessee Valley through dams, electricity, roads, canals, flood control, schools, and other public infrastructure.
- **Core empirical challenge.** The TVA region was poor and agricultural before treatment, so simple comparisons with the rest of the United States would confound the policy effect with preexisting regional differences and convergence trends.
- **Identification idea.** The authors compare TVA counties to similar untreated counties, especially counties in proposed regional authorities that Congress seriously considered but never approved. These proposed authorities were meant to be “little TVAs” and therefore form a historically motivated counterfactual.
- **Main local result.** TVA accelerated the region’s shift from agriculture to manufacturing. Agricultural employment gains disappear after federal subsidies end, but manufacturing employment continues growing faster for decades.
- **Interpretation of persistence.** The different post-subsidy trajectories of agriculture and manufacturing suggest that manufacturing exhibits agglomeration economies, while agriculture does not.
- **National-welfare question.** Local job gains are not enough to justify place-based policy if gains in the treated region are offset by losses elsewhere. The paper therefore builds a spatial equilibrium model to separate direct infrastructure effects from indirect agglomeration effects.
- **National result.** The TVA’s direct infrastructure investments raised national manufacturing productivity enough for benefits to exceed federal costs. However, indirect agglomeration effects appear to cancel out nationally because the estimated agglomeration elasticity is roughly constant across places.
- **Economic takeaway.** TVA was locally transformative and nationally beneficial mainly because it financed productive infrastructure, not because it permanently moved the economy to a more efficient spatial allocation through agglomeration spillovers.

2 Incremental contribution of the paper

- **From short-run job creation to long-run regional development.** Many place-based evaluations ask whether subsidies create local jobs in the short run. This paper follows the regional economy for roughly 70 years after the policy’s start.
- **Historical counterfactual contribution.** The use of proposed but never implemented regional authorities is an unusually creative control strategy. It uses political history to identify places that were plausible candidates for TVA-style treatment.
- **Sector-specific evidence on persistence.** The paper separates agriculture from manufacturing. This distinction is central because agglomeration economies are expected in manufacturing but not necessarily in agriculture.

- **Bridge between reduced-form and structural evaluation.** The paper first estimates local treatment effects, then embeds those facts in a model that can speak to national welfare.
- **A practical welfare framework for place-based policy.** The structural model clarifies when local gains are national gains: reallocation raises national output only if the receiving place has a higher agglomeration elasticity or if the policy directly raises productivity.
- **Empirical test of the big-push logic.** The TVA is a large-scale historical big-push experiment. The paper asks whether a massive investment can push a poor region into a permanently higher development path.
- **Substantive policy contribution.** The paper shows that TVA-like programs can pass cost-benefit tests, but only under specific conditions. Productive infrastructure matters; pure spatial relocation of manufacturing activity has weak aggregate justification when agglomeration elasticities are constant.

3 Program background and institutional setting

3.1 Creation and objectives of the TVA

- The TVA was created by Congress on May 18, 1933, during the New Deal.
- Its service area covered 163 counties across Tennessee and parts of Kentucky, Alabama, Mississippi, and neighboring states.
- The program’s mandate was broad: electricity generation, navigation, flood control, roads, canals, schools, health services, soil conservation, reforestation, fertilizer programs, and general regional modernization.
- The most visible investments were hydroelectric dams and transmission infrastructure.
- Electricity was mostly sold inside the TVA service area, often at reduced rates to municipal power authorities and cooperatives.

Interpretation: TVA was not a narrow business tax credit. It was a very large public-capital and regional modernization program, making it close to a real-world big-push intervention.

3.2 Timing of federal transfers

- Federal appropriations to TVA between 1934 and 2000 totaled roughly \$20 billion in 2000 dollars.
- Most investment occurred between 1940 and 1958, when the dams, roads, canals, and major physical infrastructure were built.
- At the peak, around 1950-1955, the annual federal subsidy was approximately \$150 per resident, or roughly \$750 per typical household because households had about five members.
- The peak household subsidy represented about 10% of average household income.
- In 1959, Congress made the TVA power system self-financing. Federal transfers then declined sharply and remained low afterward.

Interpretation: The timing creates a powerful test of persistence. If employment gains only reflect contemporaneous subsidy spending, they should disappear after 1960. If manufacturing

agglomeration has cumulative effects, manufacturing gains may persist or deepen after subsidies lapse.

3.3 Selection of the TVA region

- TVA counties were selected because they were rural, flood-prone, under-electrified, poor in public facilities, and within feasible transmission distance of power plants.
- These selection criteria mean TVA counties were not randomly chosen.
- Before the program, TVA counties were poorer, more agricultural, less urbanized, less literate, and less industrial than average U.S. counties.
- The comparison problem is therefore not simply a missing-data issue: it is a serious selection issue.

4 Data and measurement

4.1 County panel data

- The authors construct a county-level panel covering 1900 to 2000.
- Sources include the Population Census, Manufacturing Census, Agricultural Census, and topographic data collected by Fishback, Haines, and Kantor.
- Outcomes include population, employment, housing units, manufacturing wages, agricultural and manufacturing employment, agricultural land values, housing values, family income, and farm production.
- The key unit of observation is the county.

Interpretation: County data allow a long-run regional evaluation but require careful treatment of changing definitions and data quality, especially in early Census years.

4.2 Treatment and control groups

- **Treatment group:** 163 TVA service-area counties.
- **Control group 1:** non-TVA counties in the rest of the United States.
- **Control group 2:** non-TVA counties in the South.
- **Control group 3:** counties in proposed regional authorities that were considered by Congress but never implemented.
- The proposed authorities include the Atlantic Seaboard, Great Lakes-Ohio Valley, Missouri Valley, Arkansas Valley, Columbia, and Western authorities.

Interpretation: The proposed-authority controls are especially important because they share the historical logic of TVA selection: they were places policymakers thought might need TVA-style intervention.

4.3 Trimming and balancing

- TVA counties differ from many non-TVA counties. To improve comparability, the authors estimate propensity scores for TVA assignment based on preprogram characteristics.
- When comparing TVA to the whole United States or to the South, control counties with predicted treatment probability in the bottom 25% are dropped.
- When comparing TVA to proposed authorities, the authors do not trim, because the identification strategy is based on the historical accident that the authorities were proposed but never adopted.
- Preprogram covariates include 1920 and 1930 levels and trends in population, employment, sectoral shares, wages, farm values, housing values, literacy, urbanization, racial composition, radio ownership, unemployment, elevation, and elevation range.

Interpretation: The empirical design tries to compare TVA counties not to the average U.S. county, but to counties that looked similar before the intervention.

5 Reduced-form empirical design

5.1 Oaxaca-Blinder counterfactual growth regressions

The reduced-form evaluation estimates changes in outcomes over long horizons. For non-TVA counties, the authors estimate

$$y_{it} - y_{i,t-1} = \alpha + X_i' \beta + (\varepsilon_{it} - \varepsilon_{i,t-1}), \quad (1)$$

where y_{it} is an outcome for county i in decade t and X_i is a rich vector of preprogram covariates. The estimated coefficients are then used to predict how TVA counties would have grown absent treatment.

- This is an Oaxaca-Blinder adjustment: the control counties are used to estimate how preprogram characteristics predict growth.
- The predicted counterfactual for TVA counties uses the TVA counties' own preprogram characteristics.
- The method can be interpreted as a reweighting estimator: more TVA-like control counties receive greater implicit weight.

Interpretation: The method is designed to estimate the average treatment effect on treated TVA counties, while allowing effects to be heterogeneous across counties.

5.2 Inference

- The authors report standard errors clustered by state.
- They also report spatial HAC standard errors following Conley (1999), because county-level shocks may be spatially correlated across nearby counties, including counties in different states.
- Bordering non-TVA counties are dropped to avoid direct spillovers from the TVA region contaminating the control group.

5.3 Placebo test

- The key placebo examines whether TVA counties and controls had different trends before treatment, using 1900-1940 changes.
- Because most federal TVA investment occurred after 1940, strong TVA-control differences in 1900-1940 trends would indicate selection bias.
- Before controls, some differences exist, especially in agricultural land values.
- After controlling for 1920 and 1930 covariates, placebo differences become small and statistically insignificant across comparison groups.

Interpretation: The placebo test supports the view that the covariates balance preprogram trends, though it cannot rule out all possible unobserved post-1940 shocks.

6 Reduced-form local results

6.1 Long-run effects, 1940-2000

- TVA has little statistically significant effect on total population, manufacturing wages, housing values, or agricultural land values over the full 1940-2000 period.
- TVA strongly shifts employment out of agriculture and into manufacturing.
- Conditional estimates imply that agricultural employment growth is lower in TVA counties, while manufacturing employment growth is higher.
- These findings are robust whether controls are the entire United States, the South, or proposed authorities.
- Median family income tends to rise in the TVA region, especially when comparison groups are the South or proposed authorities.

Economic message: TVA did not mainly raise local wages. Instead, it changed the composition of employment toward higher-paying manufacturing jobs.

6.2 Subperiod effects, 1940-1960 versus 1960-2000

- During 1940-1960, the period of largest federal transfers, TVA counties experience large gains in both agricultural and manufacturing employment.
- During 1960-2000, when federal transfers largely end, agricultural employment falls back relative to comparison counties.
- Manufacturing employment continues to grow faster in TVA counties even after subsidies lapse.
- The continued manufacturing gains are smaller than in the peak subsidy period but remain statistically meaningful.

Economic message: Agriculture behaves like a sector temporarily helped by public investment. Manufacturing behaves like a sector in which early investment created persistent local advantages.

6.3 Interpretation: persistence and agglomeration

- Public infrastructure did not physically vanish after 1960, but its value would depreciate over time unless maintained locally.
- The collapse of agricultural gains after 1960 suggests limited agglomeration or self-reinforcement in agriculture.
- The persistence of manufacturing gains suggests agglomeration economies or other cumulative forces in manufacturing.
- The crucial welfare question remains whether the manufacturing gains in TVA came at the cost of lower manufacturing activity elsewhere.

7 Structural model

7.1 Spatial equilibrium setup

The model treats U.S. counties as small open economies. Labor and capital are mobile across counties at decadal frequencies. Worker utility depends on wages and amenities:

$$\ln w_{it} + M_{it} = u_t, \quad (2)$$

where w_{it} is the manufacturing wage, M_{it} is local amenity value, and u_t is national reservation utility.

Manufacturing output is produced with labor, capital, and a fixed local factor:

$$Y_{it} = A_{it} K_{it}^\alpha F_i^\beta L_{it}^{1-\alpha-\beta}. \quad (3)$$

Profit maximization implies an inverse labor demand curve:

$$\ln w_{it} = C - \frac{\beta}{1-\alpha} \ln L_{it} + \frac{\beta}{1-\alpha} \ln F_i - \frac{\alpha}{1-\alpha} \ln r_t + \frac{1}{1-\alpha} \ln A_{it}. \quad (4)$$

Interpretation: The fixed factor makes local labor demand slope downward. Productivity shocks or infrastructure improvements shift labor demand outward.

7.2 Local productivity and agglomeration

Productivity is decomposed as

$$\ln A_{it} = g\left(\frac{L_{i,t-1}}{R_i}\right) + \delta_t D_i + \mu_i + \eta_t + \varepsilon_{it}, \quad (5)$$

where $L_{i,t-1}/R_i$ is lagged manufacturing employment density, D_i indicates TVA exposure, δ_t is the direct productivity effect of TVA, μ_i is fixed locational advantage, η_t is a national productivity component, and ε_{it} is an idiosyncratic county productivity component.

- The function $g(\cdot)$ captures agglomeration economies.
- A lag in agglomeration avoids contemporaneous indeterminacy and creates dynamic adjustment.
- The TVA can affect productivity directly through infrastructure and indirectly by increasing density, which feeds into agglomeration.

7.3 When can local gains raise national output?

In steady state, the indirect output effect of moving a worker from county i to county j depends on average labor productivity and the local agglomeration elasticity. Reallocation raises output if

$$\frac{Y_i}{L_i}(1 - \alpha - \beta + \theta_i) < \frac{Y_j}{L_j}(1 - \alpha - \beta + \theta_j), \quad (6)$$

where θ_i is the local agglomeration elasticity.

- If amenities and wages are equal, then average labor productivity is equal, and reallocation raises output only if the receiving county has a higher agglomeration elasticity.
- If agglomeration elasticities are constant everywhere, moving manufacturing activity across counties has no aggregate productivity effect.
- If agglomeration elasticities are strongly nonlinear, place-based policy can raise aggregate output by moving workers into places where spillovers are especially strong.

Interpretation: Agglomeration can be a local market failure without being an aggregate reason for federal spatial redistribution.

8 Structural estimation and national cost-benefit analysis

8.1 Estimating equation

The authors approximate the unknown agglomeration function $g(\cdot)$ with a three-piece spline in manufacturing density:

$$g\left(\frac{L_{it}}{R_i}; \theta\right) = \sum_{k=1}^3 \theta_k g_k\left(\frac{L_{it}}{R_i}\right). \quad (7)$$

They estimate a differenced labor-demand equation:

$$\begin{aligned} \ln L_{it} - \ln L_{i,t-1} = & -\frac{1-\alpha}{\beta} (\ln w_{it} - \ln w_{i,t-1}) + \frac{\delta_t - \delta_{t-1}}{\beta} D_i \\ & + \sum_{k=1}^3 \frac{\theta_k}{\beta} \left[g_k\left(\frac{L_{i,t-1}}{R_i}\right) - g_k\left(\frac{L_{i,t-2}}{R_i}\right) \right] + X_i' \tilde{\gamma} + \tilde{\eta}_t - \tilde{\eta}_{t-1} + \tilde{\xi}_{it}. \end{aligned} \quad (8)$$

Interpretation: The model explains manufacturing employment growth using wage growth, lagged changes in manufacturing density, TVA direct effects, and baseline controls.

8.2 Identification of agglomeration

- OLS may confound true agglomeration feedback with serially correlated productivity shocks.
- The authors instrument changes in lagged density spline components using twice-lagged changes in those same spline components.
- The exclusion restriction requires productivity shocks not to remain correlated over a 20-year horizon after controls.
- The controls include extensive 1920 and 1930 county characteristics; robustness checks also add 1940 manufacturing density and regional trends.

- The short-run labor demand elasticity is calibrated at 1.5 based on Hamermesh’s review of labor demand evidence.

Interpretation: The identifying variation comes from earlier shifts in manufacturing density that should predict later growth through agglomeration, not through very long-lived unobserved shocks.

8.3 Structural results

- When the agglomeration function is modeled as linear in density levels, the marginal effect of an additional manufacturing worker per square mile is much larger in low-density counties than in high-density counties.
- This level-based pattern is strongly concave and might appear to support targeting low-density places.
- However, the relevant welfare object is the elasticity, not the marginal effect in levels.
- When the function is modeled in logs, the estimated agglomeration elasticity is similar across low-, medium-, and high-density counties.
- The IV estimates imply an elasticity of labor demand with respect to lagged manufacturing density of roughly 0.40-0.46 across all density ranges.
- The authors cannot reject equal elasticities across density groups.

Economic message: Agglomeration exists, but it is approximately constant-elasticity. This means TVA’s local spillover gains are offset by spillover losses elsewhere.

8.4 Direct productivity effects and cost-benefit results

- The direct TVA effect on labor demand is large in 1940-1960 and close to zero or slightly negative afterward.
- Using model assumptions, the 1940-1960 effect implies an approximately 8.7% increase in local manufacturing productivity.
- The authors estimate that TVA raised national manufacturing productivity by about 0.3%.
- The present value of long-run benefits is estimated at \$23.8-36.5 billion, depending on labor-supply assumptions.
- The present value of federal transfers is estimated at \$17.3 billion.
- The positive net benefit is explained by direct infrastructure productivity gains, not by aggregate agglomeration gains.

Interpretation: TVA passes the paper’s cost-benefit test because infrastructure had real productivity value. It does not pass because the government successfully exploited a nonlinear agglomeration market failure.

9 Tables and figures

9.1 Figure I and Figure II

Read:

- Figure I maps the TVA service area as of 2010. The region covers much of Tennessee and extends into Kentucky, Alabama, Mississippi, and neighboring states.
- Figure II plots annual federal transfers to TVA in 2000 dollars, both total and per resident.
- Transfers are concentrated between 1940 and the late 1950s, with peaks during World War II and the early 1950s.
- After Congress made the TVA power system self-financing in 1959, transfers fall sharply.

Economic message: The program has a clear high-subsidy period and a later low-subsidy period. This timing lets the authors distinguish contemporaneous subsidy effects from persistent regional transformation.

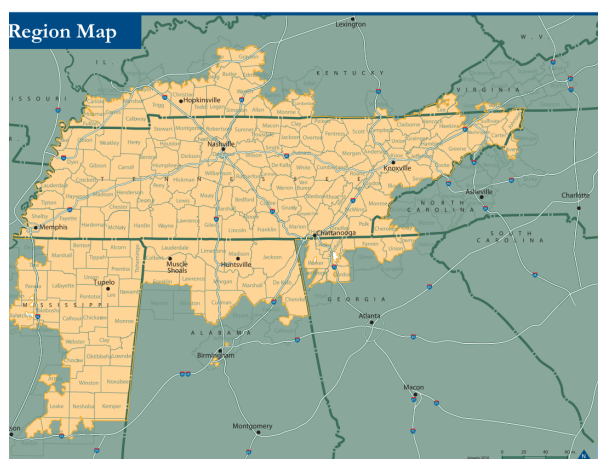


Figure I: TVA service area

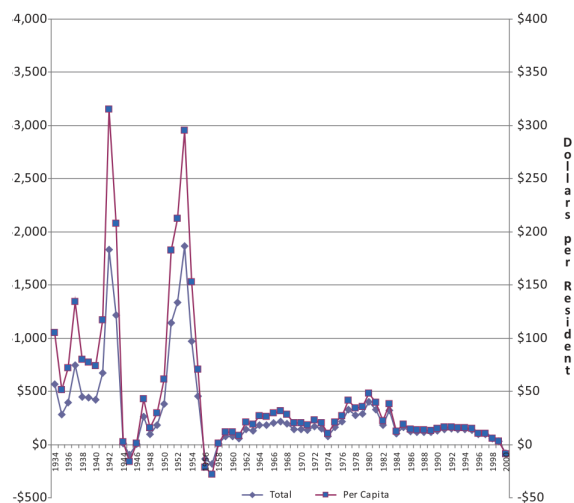


Figure II: Federal transfers to TVA

9.2 Table I: Summary statistics

Read:

- Table I compares TVA counties to non-TVA counties, non-TVA Southern counties, proposed-authority counties, and trimmed control samples.
- In 1930, TVA counties are more agricultural and less manufacturing-intensive than the typical U.S. county.
- TVA counties have lower manufacturing wages, farm values, housing values, urbanization, literacy, and radio ownership.
- Proposed-authority counties look more similar to TVA counties than the average non-TVA county.
- The lower panel shows 1920-1930 changes, demonstrating that trends also differ across regions before the program.

Economic message: TVA counties were not randomly selected. Any credible evaluation must condition on preprogram characteristics and trends.

	(1)	(2)	(3)	(4)	(5)	(6)
	Overall		Trimmed sample		Trimmed sample	
	TVA	Non-TVA	Non-TVA South	Non-TVA proposed authorities	Non-TVA	Non-TVA South
1930 characteristics						
Log population	9.991	9.977	9.989	9.940	9.905	9.979
Log employment	8.942	8.967	8.959	8.908	8.881	8.947
Log # of houses	8.445	8.508	8.455	8.466	8.442	8.445
Log average manufacturing wage	1.406	1.802	1.545	1.685	1.728	1.538
Manufacturing employment share	0.075	0.090	0.080	0.077	0.080	0.078
Agricultural employment share	0.017	0.455	0.541	0.510	0.487	0.547
% White	0.813	0.885	0.722	0.830	0.863	0.724
% Urbanized	0.153	0.280	0.233	0.216	0.242	0.215
% Illiterate	0.088	0.045	0.092	0.060	0.051	0.092
% of Whites foreign born	0.002	0.059	0.013	0.020	0.030	0.011
Log average farm value	5.252	5.646	5.386	5.552	5.579	5.370
Log median housing value	9.271	9.581	9.360	9.432	9.516	9.308
Log median contract rent	8.574	9.030	8.679	8.834	8.894	8.672
% Own radio	0.079	0.296	0.114	0.210	0.256	0.112
Max elevation (meters)	1,576.190	2,364.531	1,068.943	1,758.893	2,044.656	1,070.334
Elevation range (max-min)	1,127.761	1,521.322	712.536	1,083.293	1,251.074	715.253
% Counties in South	1.000	0.342	1.000	0.554	0.447	1.000

Table I, part 1: 1930 levels

	(1)	(2)	(3)	(4)	(5)	(6)
	Overall		Trimmed sample		Trimmed sample	
	TVA	Non-TVA	Non-TVA South	Non-TVA proposed authorities	Non-TVA	Non-TVA South
Changes 1920-1930						
Log population	0.051	0.049	0.067	0.004	0.037	0.060
Log employment	0.082	0.096	0.111	0.045	0.083	0.103
Log # of houses	0.078	0.092	0.108	0.046	0.078	0.100
Log average manufacturing wage	0.117	0.217	0.108	0.172	0.197	0.103
Manufacturing employment share	-0.010	-0.035	-0.018	-0.018	-0.026	-0.018
Agricultural employment share	-0.047	-0.036	-0.047	-0.046	-0.042	-0.047
% White	0.012	-0.011	-0.010	0.000	-0.006	-0.004
% Urbanized	0.047	0.064	0.080	0.042	0.054	0.069
% Illiterate	-0.030	-0.014	-0.029	-0.019	-0.015	-0.028
% of Whites foreign born	-0.001	-0.023	-0.016	-0.012	-0.015	-0.012
Log average farm value	-0.013	-0.076	0.025	-0.182	-0.102	0.013
# of Observations	163	2,326	795	828	1744	779
# of States	6	46	14	25	43	14

Table I, part 2: 1920-1930 changes

9.3 Figure III and Table II

Read:

- Figure III shows the implicit weights assigned to untreated counties by the Oaxaca-Blinder estimator.
- Darker untreated counties receive more weight because their preprogram characteristics make them more similar to TVA counties.
- Table II implements the placebo test: it estimates false TVA effects on 1900-1940 preprogram trends.
- Before controls, some outcomes differ between TVA counties and the full U.S. control sample.
- After controls, placebo estimates are small and statistically insignificant across the three counterfactual groups.

Economic message: The comparison strategy does not rely on the average untreated county. It constructs a TVA-like counterfactual, and the placebo test supports its credibility.

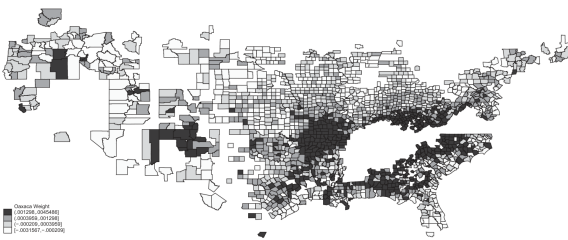


Figure III: Oaxaca-Blinder weights on untreated counties

Outcome	(1) Point estimate (unadjusted)	(2) Clustered std. err.	(3) Point estimate (controls)	(4) Clustered std. err.	(5) Spatial HAC	(6) N
Panel A: TVA region versus rest of U.S.						
Population	0.007	(0.016)	0.010	(0.012)	(0.016)	1,776
Total employment	-0.009	(0.016)	0.005	(0.013)	(0.016)	1,776
Housing units	-0.006	(0.015)	0.007	(0.011)	(0.013)	1,776
Average manufacturing wage	0.009	(0.018)	0.010	(0.021)	(0.016)	1,428
Manufacturing share	0.007*	(0.004)	0.005	(0.004)	(0.005)	1,776
Agricultural share	-0.007*	(0.004)	-0.001	(0.005)	(0.005)	1,776
Average agricultural land value	0.078***	(0.021)	0.025	(0.018)	(0.018)	1,746
Panel B: TVA region versus U.S. South						
Population	-0.018	(0.018)	0.003		(0.016)	850
Total employment	-0.028	(0.018)	0.001		(0.016)	850
Housing units	-0.025	(0.016)	0.005		(0.013)	850
Average manufacturing wage	0.001	(0.015)	0.001		(0.016)	687
Manufacturing share	0.005	(0.005)	0.005		(0.005)	850
Agricultural share	0.003	(0.004)	-0.002		(0.005)	850
Average agricultural land value	-0.009	(0.020)	-0.007		(0.017)	839

Table II, part 1: placebo estimates, panels A and B

9.4 Table II continued and Table III, part 1

Read:

- The continued part of Table II reports the placebo comparison with proposed authorities.
- Proposed authorities again show no significant conditional pretrend differences after controls.
- Table III begins the main long-run reduced-form estimates for 1940-2000.

- Relative to the full United States and the South, TVA counties show large reductions in agricultural employment growth and large increases in manufacturing employment growth after controls.

Economic message: The same design that passes the placebo test finds large post-treatment sectoral shifts, strengthening the interpretation that those shifts are policy-related.

TABLE II
(CONTINUED)

Outcome	(1) Point estimate (unadjusted)	(2) Clustered std. err.	(3) Point estimate (controls)	(4) Clustered std. err.	(5) Spatial HAC	(6) N
Panel C: TVA region versus proposed authorities						
Population	0.028	(0.019)	0.011		(0.016)	926
Average manufacturing wage	-0.012	(0.017)	0.006		(0.015)	926
Agricultural employment	-0.014	(0.016)	0.006		(0.013)	926
Value of farm production	0.012	(0.015)	0.008		(0.017)	734
Median family income (1950-2000 only)	0.007	(0.006)	0.005		(0.006)	926
Average agricultural land value	-0.005	(0.006)	0.004		(0.006)	926
Median housing value	0.080***	(0.026)	0.017		(0.018)	908

Notes: Column (1) gives the unconditional difference between TVA and non-TVA counties in the 1900-1940 change in the log of the relevant outcome divided by 4 (share not converted to logs). Column (3) adjusts for preprogram differences between TVA counties and controls via a Oaxaca-Blinder regression as in Kling (2011). Covariates include time-invariant county, regional, and national fixed effects, time-invariant county, regional, and national fixed effects, and demographic characteristics (see Section III A for full list of covariates).

ECONOMIC EFFECTS OF THE TVA

Table II, part 2: proposed-authority placebo

Outcome	(1) Point estimate (unadjusted)	(2) Clustered std. err.	(3) Point estimate (controls)	(4) Clustered std. err.	(5) Spatial HAC	(6) N
Panel A: TVA region versus rest of U.S.						
Population	0.004	(0.021)	0.007	(0.020)	(0.018)	1,907
Average manufacturing wage	0.027***	(0.006)	0.005	(0.004)	(0.005)	1,172
Agricultural employment	-0.130***	(0.026)	-0.056**	(0.024)	(0.027)	1,907
Manufacturing employment	0.076***	(0.013)	0.059***	(0.015)	(0.023)	1,907
Value of farm production	-0.028	(0.028)	0.002	(0.032)	(0.026)	1,903
Median family income (1950-2000 only)	0.072***	(0.014)	0.021	(0.013)	(0.011)	1,905
Average agricultural land value	0.066***	(0.013)	-0.002	(0.012)	(0.016)	1,906
Median housing value	0.040**	(0.017)	0.005	(0.015)	(0.015)	1,906
Panel B: TVA region versus U.S. South						
Population	-0.007	(0.018)	0.014		(0.019)	942
Average manufacturing wage	0.003	(0.006)	0.001		(0.005)	610
Agricultural employment	-0.097***	(0.030)	-0.051*		(0.027)	942
Manufacturing employment	0.079***	(0.023)	0.063***		(0.024)	942
Value of farm production	-0.005	(0.025)	-0.006		(0.026)	939
Median family income (1950-2000 only)	0.041***	(0.012)	0.024**		(0.011)	942
Average agricultural land value	0.031*	(0.018)	-0.003		(0.017)	942
Median housing value	0.019	(0.017)	0.007		(0.016)	942

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Table III, part 1: long-run TVA effects

9.5 Table III continued and Table IV

Read:

- Table III continued reports long-run estimates using proposed authorities as the comparison group.
- The estimated long-run effect on manufacturing employment is positive and significant; the estimated effect on agricultural employment is negative.
- Table IV separates effects into 1940-1960 and 1960-2000.
- In 1940-1960, TVA raises both agricultural and manufacturing employment growth.
- In 1960-2000, agricultural employment growth reverses, while manufacturing employment growth remains positive.

Economic message: TVA's short-run effects are broad, but only manufacturing exhibits post-subsidy persistence.

TABLE III
(CONTINUED)

Outcome	(1) Point estimate (unadjusted)	(2) Clustered std. err.	(3) Point estimate (controls)	(4) Clustered std. err.	(5) Spatial HAC	(6) N
Panel C: TVA region versus proposed authorities						
Population	0.011	(0.018)	0.001		(0.017)	991
Average manufacturing wage	0.018***	(0.007)	0.005		(0.006)	618
Agricultural employment	-0.101***	(0.029)	-0.071***		(0.027)	991
Manufacturing employment	0.066***	(0.024)	0.053**		(0.024)	991
Value of farm production	0.002	(0.026)	0.011		(0.035)	989
Median family income (1950-2000 only)	0.060***	(0.012)	0.025**		(0.011)	991
Average agricultural land value	0.060***	(0.019)	-0.003		(0.016)	991
Median housing value	0.033**	(0.016)	0.009		(0.016)	991

Notes: Point estimates obtained from regression of 1940-2000 change in outcomes divided by 6 on TVA dummy. All outcomes besides share variables are transformed to changes in log outcomes. Asterisks indicate significance at the 10% level, ** significant at 5% level, *** significant at 1% level.

ECONOMIC EFFECTS OF THE TVA

Table III, part 2: long-run estimates with proposed authorities

SPATIALLY VARYING IMPACT OF TVA ON GROWTH RATE OF OUTCOMES OVER TWO SUBPERIODS

Outcome	(1) Entire U.S.	(2) 1940-1960	(3) 1940-1960	(4) 1960-2000	(5) Proposed authorities	(6) 1940-1960	(7) 1960-2000
Population	0.037	-0.008	0.042	-0.000	0.028	-0.013	-0.013
Average manufacturing wage	-0.005	0.014*	-0.003	0.010	0.007	0.012	-0.160***
Agricultural employment	0.106***	-0.134***	0.106***	-0.130***	0.119***	-0.160***	-0.160***
Manufacturing employment	0.114***	0.033**	0.116***	0.033**	0.097**	0.032**	0.032**
Value of farm production	0.076*	-0.030	0.081**	-0.044	0.118**	-0.033	-0.033
Median family income	N/A	0.017	N/A	0.016	N/A	0.019*	0.019*
Average agricultural land value	0.027	-0.017	0.018	-0.015	0.029	-0.021	-0.021
Median housing value	0.019	-0.003	0.010	0.005	0.020	0.003	0.003

Notes: Full set of controls included in all specifications. Point estimates obtained from Oaxaca-Blinder regression of 1940-1960 or 1960-2000 change in log outcomes divided by 6 on TVA dummy and interacted controls as in Kling (2011). Covariates include time-invariant geographic characteristics and trends in program district mix, population, and demographic characteristics (see Section III A for full list of covariates). Asterisks based on standard errors clustered by state (entire U.S.) or spatial AC estimates (South and Proposed Authorities) using technique of Cooley (1986) with bandwidth of 200 miles. * significant at 10% level, ** significant at 5% level, *** significant at 1% level.

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Table IV: effects in 1940-1960 and 1960-2000

9.6 Figure IV: Dynamics under linear and nonlinear agglomeration

Read:

- Panel A shows dynamics with a log-linear agglomeration function. A TVA productivity shift moves the region gradually toward a new steady state, but with depreciation it eventually returns toward its old position.
- Panel B shows nonlinear agglomeration with a poverty trap and a developed steady state. If a big push crosses the tipping point, effects can become permanent.
- The figure makes clear why the shape of the agglomeration function is central to welfare analysis.

Economic message: Persistence alone does not prove a permanent big-push effect. Slow convergence under linear agglomeration can also generate long-lasting effects.

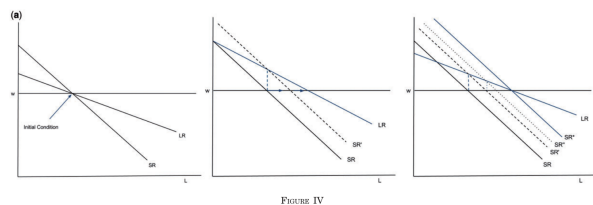


Figure IV, Panel A: linear agglomeration dynamics

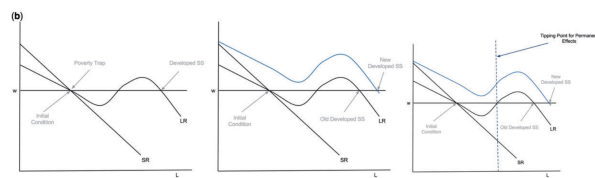


Figure IV, Panel B: nonlinear agglomeration dynamics

9.7 Tables V and VI: Structural agglomeration estimates

Read:

- Table V estimates the agglomeration function using a linear basis in density levels.
- The estimates imply strong concavity in levels: adding one worker per square mile matters more in low-density manufacturing counties.
- Table VI estimates the agglomeration function using a log basis.
- In the log specification, low-, medium-, and high-density areas have very similar estimated agglomeration elasticities.
- The paper therefore cannot reject constant elasticities across density ranges.

Economic message: The level specification shows concavity, but the policy-relevant elasticity is roughly constant. This is the key result behind the conclusion that agglomeration gains cancel out nationally.

TABLE V
STRUCTURAL ESTIMATES OF AGGLOMERATION FUNCTION (LINEAR BASIS)

	(1) OLS	(2) OLS	(3) OLS	(4) 2SLS	(5) 2SLS	(6) 2SLS
<i>Change in manufacturing density spline components:</i>						
Low	0.066 (0.010)	0.060 (0.010)	0.060 (0.010)	0.097 (0.037)	0.084 (0.037)	0.082 (0.037)
				[129.06]	[122.25]	[121.07]
Medium	0.021 (0.0045)	0.022 (0.0044)	0.022 (0.0044)	0.042 (0.011)	0.043 (0.011)	0.042 (0.011)
				[116.87]	[114.87]	[116.66]
High	-0.000075 (0.000074)	0.000031 (0.000075)	0.00011 (0.00071)	0.0019 (0.0020)	0.0021 (0.0020)	0.0018 (0.0020)
				[41.82]	[40.96]	[32.04]
Log manufacturing wages	-1.5 0.0033 (0.015)	-1.5 0.0081 (0.015)	-1.5 0.0026 (0.015)	-1.5 -0.0052 (0.017)	-1.5 0.0012 (0.016)	-1.5 -0.0043 (0.016)
TVA						
Regional trends	no	no	yes	no	no	yes
1940 manufacturing density	no	yes	yes	no	yes	yes
Decade effects	yes	yes	yes	yes	yes	yes
Controls for 1920 and 1930 characteristics	yes	yes	yes	yes	yes	yes
<i>p</i> -value equal slopes	0.0000	0.0000	0.0000	0.0007	0.0006	0.0006
<i>p</i> -value slopes equal 0	0.0000	0.0000	0.0000	0.0022	0.0015	0.0019
<i>N</i>	6,057	6,057	6,057	5,952	5,952	5,952

Notes. Dependent variable is change in log county manufacturing employment. Manufacturing density is manufacturing employment per square mile. Standard errors clustered by state in parentheses. Angrist-Pischke cluster robust first stage *F*-statistic in brackets. All estimates weighted by 1950 county population. "Low" refers to spline component corresponding to density below 60th percentile of 1980 distribution, "Medium" to density between 60th and 85th percentile of 1980 distribution, and "High" to density above 85th percentile of 1980 distribution. Spline coefficients give the proportional effect on labor demand of an increase in lagged manufacturing density of one worker per square mile over the relevant range. The instruments are changes in the spline components of manufacturing density lagged by two decades.

Table V: structural estimates, linear density basis

TABLE VI
STRUCTURAL ESTIMATES OF AGGLOMERATION FUNCTION (LOG BASIS)

	(1) OLS	(2) OLS	(3) OLS	(4) 2SLS	(5) 2SLS	(6) 2SLS
<i>Change in log manufacturing density spline components:</i>						
Low	0.173 (0.037)	0.147 (0.037)	0.146 (0.037)	0.443 (0.102)	0.400 (0.108)	0.396 (0.107)
				[177.17]	[159.14]	[157.20]
Medium	0.221 (0.045)	0.227 (0.044)	0.226 (0.045)	0.456 (0.124)	0.440 (0.123)	0.438 (0.124)
				[106.74]	[109.55]	[110.13]
High	0.143 (0.051)	0.012 (0.050)	0.141 (0.050)	0.466 (0.150)	0.467 (0.150)	0.453 (0.151)
				[206.66]	[204.69]	[200.36]
Log manufacturing wages	-1.5 0.007 (0.014)	-1.5 0.012 (0.014)	-1.5 0.008 (0.014)	-1.5 -0.003 (0.012)	-1.5 0.002 (0.013)	-1.5 -0.002 (0.012)
TVA						
Regional trends	no	no	yes	no	no	yes
1940 manufacturing density	no	yes	yes	no	yes	yes
Decade effects	yes	yes	yes	yes	yes	yes
Controls for 1920 and 1930 characteristics	yes	yes	yes	yes	yes	yes
<i>p</i> -value equal slopes	0.2483	0.1298	0.1038	0.9545	0.6695	0.7171
<i>p</i> -value slopes equal 0	0.0000	0.0000	0.0000	0.0002	0.0007	0.0016
<i>N</i>	6,057	6,057	6,057	5,935	5,935	5,935

Notes. Dependent variable is change in log county manufacturing employment. Manufacturing density is manufacturing employment per square mile. Standard errors clustered by state in parentheses. Angrist-Pischke cluster robust first stage *F*-statistic in brackets. All estimates weighted by 1950 county population. "Low" refers to spline component corresponding to log density below 60th percentile of 1980 distribution, "Medium" to log density between 60th and 85th percentile of 1980 distribution, and "High" to log density above 85th percentile of 1980 distribution. Spline coefficients give the elasticity of labor demand with respect to lagged manufacturing density over the relevant range. The instruments are changes in the spline components of log manufacturing density lagged by two decades.

Table VI: structural estimates, log density basis

9.8 Table VII and Figure V

Read:

- Table VII estimates direct TVA effects on labor demand by period.
- The direct effect is large in 1940-1960 and negative but statistically insignificant after 1960.
- Figure V plots the calibrated short-run inverse labor demand curve and the estimated long-run inverse labor demand curve.
- The estimated long-run curve is flatter than the short-run curve, consistent with agglomeration reducing the effect of fixed factors.
- The shape implies a unique steady state, not a permanent multiple-equilibrium big-push transition.

Economic message: TVA had a large direct productivity effect during the investment period. The structural estimates imply eventual reversion rather than permanent escape from a poverty trap.

	(1) 1940–1960	(2) 1960–1980	(3) 1980–2000
Spline in levels	0.225 (0.070)	−0.011 (0.041)	−0.004 (0.039)
Spline in logs	0.185 (0.081)	−0.036 (0.033)	−0.033 (0.035)
Controls for 1920 and 1930 characteristics	yes	yes	yes
<i>N</i>	1,587	1,498	1,533

Notes. Dependent variable is residualized change in log county employment over specified horizon (see text for details). Coefficients obtained from regression of residual on TVA dummy and baseline controls. “Spline in levels” specification forms residual assuming agglomeration function is three-piece spline in levels with coefficients from column (4) of Table V. “Spline in logs” specification assumes agglomeration function is three-piece spline in log of manufacturing density with coefficients from column (4) of Table VI. Standard errors clustered by state.

Table VII: direct TVA effects by period

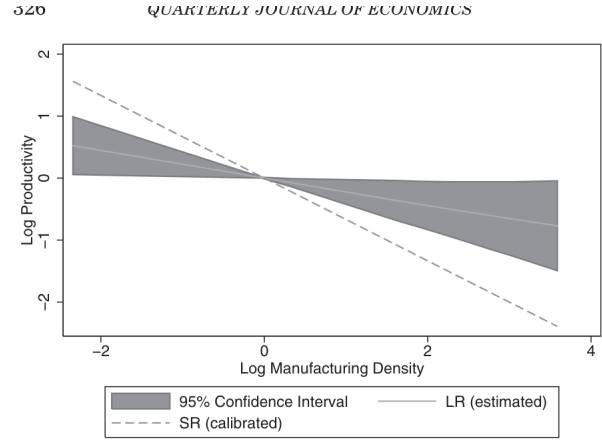


Figure V: short- and long-run inverse labor demand

10 Short summary

- The paper evaluates the Tennessee Valley Authority, one of the largest place-based development programs in U.S. history.
- TVA counties were poor, agricultural, and less industrial before treatment.
- Proposed but unapproved regional authorities provide a historically motivated comparison group.
- Reduced-form estimates show TVA shifted employment from agriculture to manufacturing.
- Agricultural gains disappear after federal subsidies end, but manufacturing gains persist.
- Persistence in manufacturing is consistent with agglomeration economies.
- The structural model shows that agglomeration creates national gains only if agglomeration elasticities are higher in receiving places.
- The estimated agglomeration elasticity is roughly constant across places.
- Therefore, TVA’s agglomeration gains in the treated region are offset by agglomeration losses elsewhere.
- TVA’s national benefits come mainly from direct productivity gains due to infrastructure.
- The program appears to pass the paper’s cost-benefit test: benefits exceed federal transfers.

11 Conclusion

11.1 Core conclusion

Kline and Moretti show that the TVA had large and persistent local effects. It sped the industrialization of the Tennessee Valley and raised regional income by replacing low-paying agricultural jobs with higher-paying manufacturing jobs. However, the national welfare interpretation is more

subtle. Local manufacturing persistence is not enough to prove that the country gained from spatial reallocation. The structural estimates indicate that the national gains came mostly from direct infrastructure-induced productivity improvements, not from moving manufacturing employment into a place with unusually high agglomeration returns.

11.2 Economic intuition

The local intuition is straightforward: infrastructure lowered the cost of doing business in a poor region, attracted manufacturing, and manufacturing then became self-reinforcing for a long time. The national intuition is different. If agglomeration spillovers have the same elasticity everywhere, then adding workers to TVA creates spillovers there but removes workers from other places where spillovers also matter. The gains and losses offset. In that case, a place-based policy improves national welfare only if it directly raises productivity or moves activity toward places with stronger agglomeration elasticities.

11.3 Incremental contribution revisited

The paper advances the literature in three connected ways. First, it provides a credible long-run evaluation of a major U.S. place-based development policy. Second, it uses historical policy proposals to form a counterfactual control group. Third, it develops a structural framework that turns local reduced-form evidence into national cost-benefit evidence. This combination lets the paper move beyond the usual question, “Did the treated region gain?” toward the harder question, “Did the country gain?”

11.4 Policy implications

The findings caution against simple enthusiasm for place-based subsidies. A locality may gain manufacturing jobs, but those jobs may come from other localities. Agglomeration externalities do not automatically justify federal relocation policy. The stronger case for TVA is that it financed productive public infrastructure in an underdeveloped region. For modern policy, the lesson is that the social value of place-based programs depends on the type of investment, the shape of agglomeration economies, and the opportunity cost of displaced activity elsewhere.

11.5 Limitations

The analysis depends on historical data with measurement limitations, especially in early county-level manufacturing and agricultural records. The reduced-form estimates cannot fully rule out unobserved post-1940 shocks unique to the TVA region. The structural welfare analysis relies on assumptions about labor mobility, capital mobility, homogeneous workers, the calibrated labor demand elasticity, and the exclusion restriction for lagged manufacturing-density instruments. The conclusions apply most directly to mid-twentieth-century U.S. manufacturing and may not generalize to modern high-tech clusters, services, or policies with different financing mechanisms.

11.6 Takeaway

The enduring takeaway is that the TVA was a successful local development policy and likely a positive national investment, but not because it generated free aggregate gains through manufacturing

spillovers. Its durable local transformation was real; its national payoff came mainly from productive infrastructure. Agglomeration economies mattered locally, but because the estimated elasticity was roughly constant across places, the indirect spillover effects washed out in the aggregate.